

Gayed-Bilello Leverage Rotation Strategy —

Personal Trade Review

Reference: *Leverage for the Long Run* — Michael A. Gayed CFA & Charlie Bilello CMT, 2016 Charles H. Dow Award

The Rule (Exact Paper Definition)

When the S&P 500 Index closes **above** its 200-day Simple Moving Average → go **long** the leveraged ETF.

When the S&P 500 Index closes **at or below** its 200-day Simple Moving Average → rotate into **Treasury bills**.

- Signal: confirmed at end-of-day close using `adjusted_close`
- Execution: next session open (no same-day lookahead)
- Filter: SMA, not EMA. Length: 200 calendar days
- Risk-off: paper uses T-bills; this backtest uses 0% cash (conservative)

How to Read the Signal (Live Use)

Condition	Action
SPY close > SMA-200 and you are in cash	Buy leveraged ETF at next open
SPY close ≤ SMA-200 and you hold the ETF	Sell leveraged ETF at next open, move to cash
No change	Hold current position

Check the signal each day after market close. Act at the following morning's open.

Metrics Comparison — Paper vs My Backtest

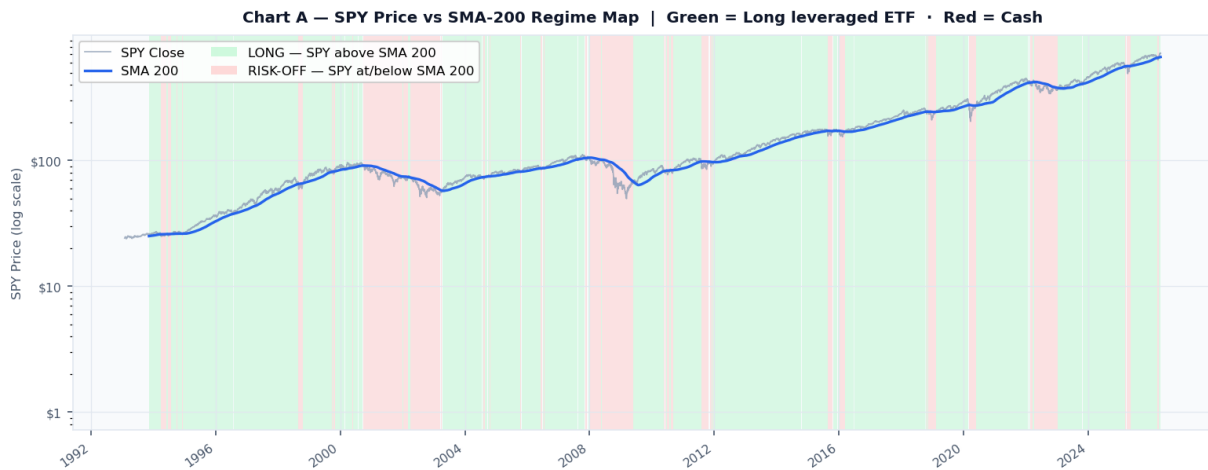
Strategy	CAGR	Ann. Vol	Sharpe	Sortino	Max DD	Trades/ Yr	Period
Paper Results — Gayed & Bilello (2016), S&P 500 synthetic leverage, Oct 1928–Oct 2015, 1% annual leverage fee, risk-off = T-bills							
S&P 500 Buy & Hold	9.1%	18.9%	0.30	0.43	−86.2%	~0	1928–2015
S&P 1.25x LRS	12.5%	15.5%	0.53	0.83	−59.0%	~5	1928–2015
S&P 2x LRS	19.1%	24.9%	0.51	0.90	−78.7%	~5	1928–2015
S&P 3x LRS	26.8%	37.3%	0.47	0.90	−92.2%	~5	1928–2015
My Backtest — SPY/SSO/UPRO ETFs, SMA-200 signal, risk-off = 0% cash, costs = 2 bps slippage+spread per side							
SPY 1x Strategy	8.1%	12.3%	0.70	0.80	−27.9%	6.6	1993-11-12 → 2026-04-24
SPY 1x Buy & Hold	10.7%	18.7%	0.64	0.82	−55.4%	0.0	1993-11-12 → 2026-04-24
SPY→SSO 2x Strategy	12.3%	23.5%	0.61	0.70	−43.5%	5.9	2006-06-21 → 2026-04-24
SSO 2x Buy & Hold	15.4%	37.8%	0.57	0.71	−84.8%	0.0	2006-06-21 → 2026-04-24
SPY→UPRO 3x Strategy	21.8%	36.6%	0.72	0.86	−58.4%	5.3	2009-06-25 → 2026-04-24
UPRO 3x Buy & Hold	32.9%	50.0%	0.82	1.03	−74.8%	0.0	2009-06-25 → 2026-04-24

Period difference: paper runs from 1928; my data starts at SPY inception (Jan 1993). SSO launched Jun 2006, UPRO launched Jun 2009. Shorter periods explain lower absolute returns on leveraged ETF strategies. Sharpe is shown without risk-free rate deduction in both paper and my backtest. Risk-off = 0% cash in my backtest vs T-bills in the paper — underestimates strategy return during high-rate periods (e.g. 2022–2024).

Charts

Chart A — SPY Regime Map

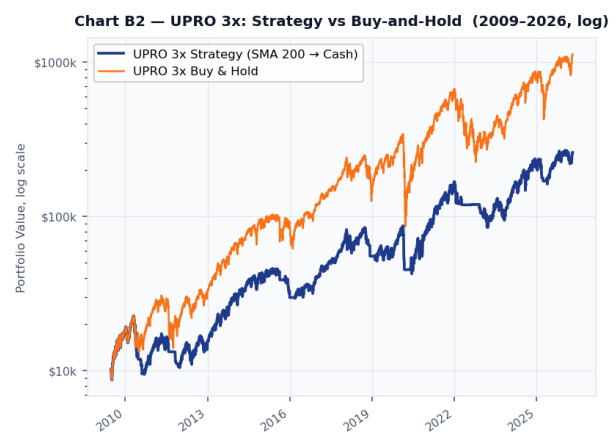
The SMA-200 divides history into LONG regimes (green) and RISK-OFF regimes (red). The signal is generated at daily close; execution happens at the next session open.



SPY price vs SMA-200. Green = long leveraged ETF. Red = move to cash.

Chart B — Growth of \$10,000

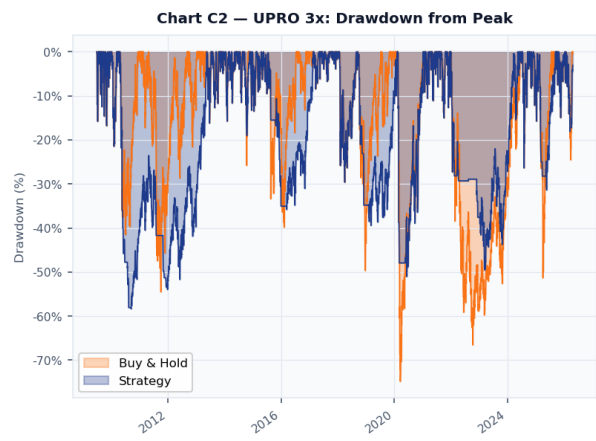
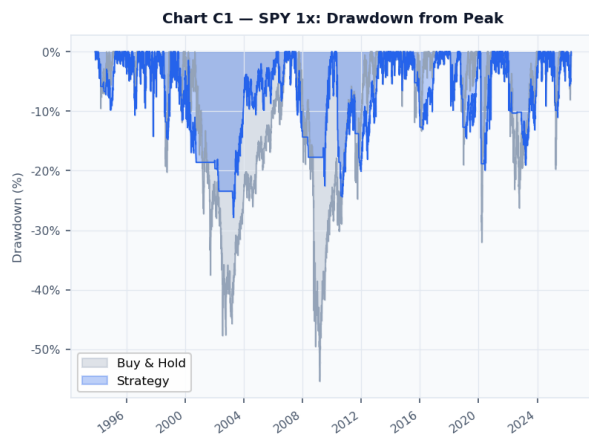
Left panel: SPY 1x — strategy vs buy-and-hold. Right panel: UPRO 3x on a log scale. The strategy exits to cash during risk-off periods, dramatically reducing the worst drawdowns of the leveraged buy-and-hold position. Both series start at \$10,000.



Growth of \$10,000. Note: UPRO data starts June 2009.

Chart C — Drawdown from Peak

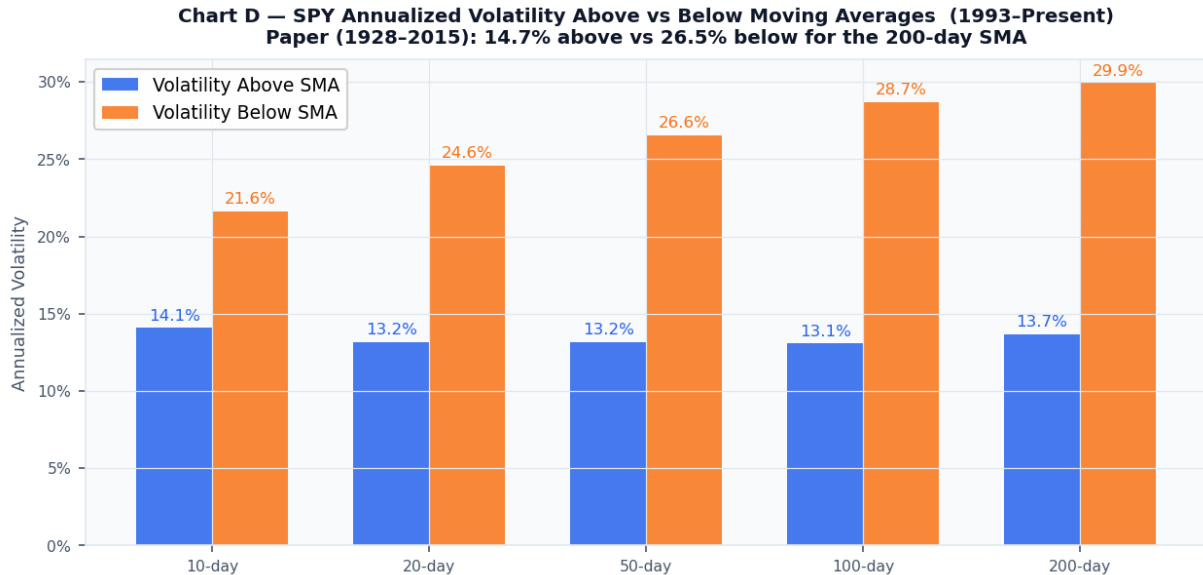
The SMA-200 timing rule's primary value is downside protection, not upside capture. The strategy significantly truncates the worst drawdowns on both SPY and UPRO.



Drawdown from peak. Shaded area shows underwater periods.

Chart D — Annualized Volatility Above vs Below SMA

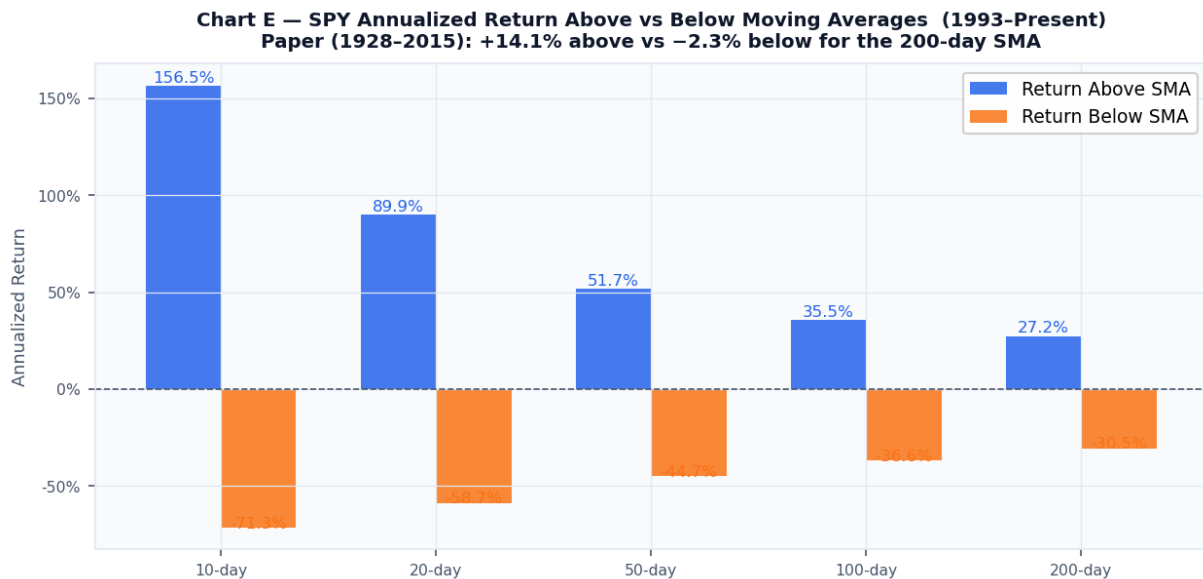
Replicates paper Chart 1 (1928–2015), now computed on SPY data from 1993. The pattern is consistent: volatility is materially lower when SPY trades above its moving average. Paper found 14.7% above vs 26.5% below for the 200-day SMA.



Annualized volatility by regime. Paper: 14.7% above vs 26.5% below (200-day, 1928–2015).

Chart E — Annualized Return Above vs Below SMA

Replicates paper Chart 3. Returns when SPY is above its SMA are strongly positive; returns when below are near zero or negative. The 200-day SMA shows the clearest separation. Paper found +14.1% above vs –2.3% below for the 200-day SMA.



Annualized return by regime. Paper: +14.1% above vs –2.3% below (200-day, 1928–2015).

Chart F — Rolling 3-Year Outperformance

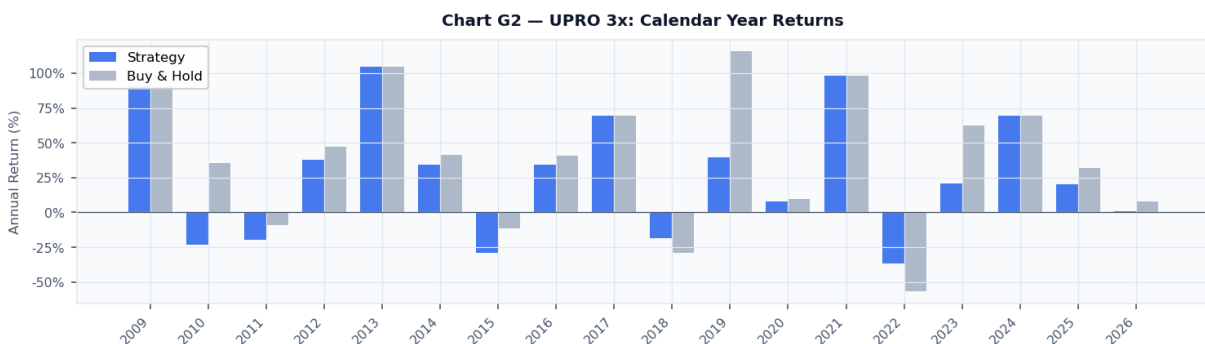
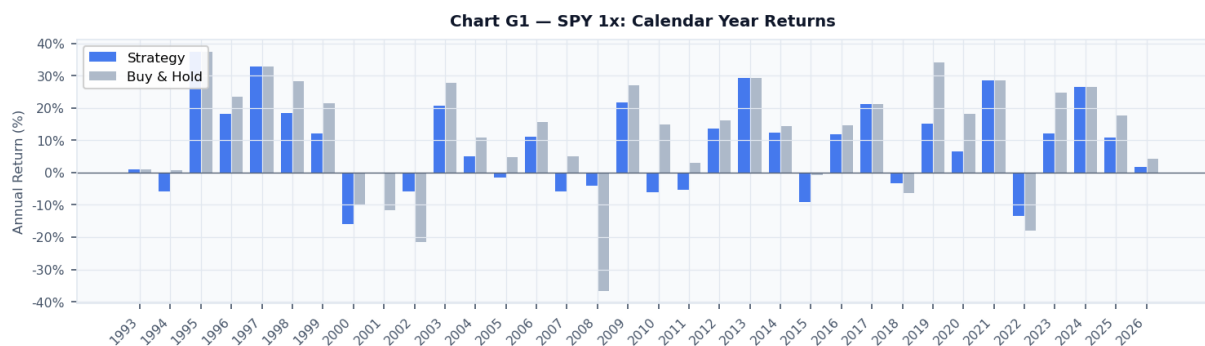
Replicates paper Chart 9. Shows how much the LRS outperforms (or underperforms) buy-and-hold on a rolling 3-year basis. Periods below zero are when the strategy lagged the market — typically strong uninterrupted bull runs (1995–2000, 2013–2021).



Rolling 3-year annualised outperformance vs buy-and-hold. Dashed = zero line.

Chart G — Calendar Year Returns

Year-by-year comparison of strategy (blue) vs buy-and-hold (gray). The strategy underperforms in strong bull years but protects capital in bear years (2000, 2002, 2008, 2022).



Calendar year returns. Strategy protects in bear years at the cost of some bull-year upside.